

Topic 17: Organic Chemistry II

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include investigating the reactions of different functional groups, identifying a carbonyl compound by preparing and finding the melting point of its 2,4-DNPH derivative, preparing and hydrolysing esters.

Mathematical skills that could be developed in this topic include representing chiral molecules with appropriate diagrams, calculating percentage yields and experimental errors.

Within this topic, students can consider how organic synthesis can produce a variety of important materials, such as esters for solvents, flavourings and perfumes. They will also continue their study of reaction mechanisms, and see the ways in which different mechanisms act as a pattern to describe a range of organic reactions.

Students should:
Topic 17A: Chirality
1. know that optical isomerism is a result of chirality in molecules with a single chiral centre
2. understand that optical isomerism results from chiral centre(s) in a molecule with asymmetric carbon atom(s) and that optical isomers are object and non-superimposable mirror images
3. know that optical activity is the ability of a single optical isomer to rotate the plane of polarisation of plane-polarised monochromatic light in molecules containing a single chiral centre
4. understand the nature of a racemic mixture
5. be able to use data on optical activity of reactants and products as evidence for S_N1 and S_N2 mechanisms